WINDOW'S POWERSHELL

22 March 2026
08:59

Windows PowerShell is used fir automation. It is more advance CLI as we can prepare payloads , scripts etc

Host Environment in PowerShell

1. CLI/ Shell - used for simple one line commands or testing power shell.
2. Development Environments(GUI with advance Capabilities) - used for complex scripts (recommended). It has build in debugger, supports differentiate language, updated and maintained by team of experts at Microsoft
3. PowerShell - preinstalled in VS code where we need to download powerful extension of Microsoft.
   PowerShell is a not case-sensitive

Command and Syntax

The command and syntax followed by PowerShell is **Verb - Noun**
Example- get -service -name "Bits"
Name : parameter and Bits: value
The command
Get - command -name "Get-Service" -syntax
or
Get-command "get-service" -syntax

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\lenovo> get-command -name "get-service" -syntax

Get-Service [[-Name] <string[]>] [-ComputerName <string[]>] [-DependentServices] [-RequiredServices] [-Include <string[]>] [-Exclude <string[]>] [<CommonPara

Get-Service -DisplayName <string[]> [-ComputerName <string[]>] [-DependentServices] [-RequiredServices] [-Include <string[]>] [-Exclude <string[]>] [<Common

Get-Service [-ComputerName <string[]>] [-DependentServices] [-RequiredServices] [-Include <string[]>] [-Exclude <string[]>] [-InputObject <ServiceController
```

Here the output as shown is multiple parameters
The parameters used in square brackets - is optional
Its not necessary to write name followed by parameter - we can write only value without name.
[-computername <string[]>] here the string is multiple values of same type.
[-DependentServices] - here only one value can be entered.

To get multiple parameters automatically - get-service "enter tab key"

Add Prefix to Search in command
Here we add prefix such as service to get all names related to word that ends with service
Get-command - noun "service"

```
PS C:\Users\lenovo> get-command -noun "service"

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Cmdlet          Get-Service                                        3.1.0.0    Microsoft.PowerShell.Management
Cmdlet          New-Service                                        3.1.0.0    Microsoft.PowerShell.Management
Cmdlet          Restart-Service                                    3.1.0.0    Microsoft.PowerShell.Management
Cmdlet          Resume-Service                                     3.1.0.0    Microsoft.PowerShell.Management
Cmdlet          Set-Service                                        3.1.0.0    Microsoft.PowerShell.Management
Cmdlet          Start-Service                                      3.1.0.0    Microsoft.PowerShell.Management
Cmdlet          Stop-Service                                       3.1.0.0    Microsoft.PowerShell.Management
Cmdlet          Suspend-Service                                    3.1.0.0    Microsoft.PowerShell.Management

PS C:\Users\lenovo> get-command -verb "copy"
```

Add Suffix to search in Command
Here we add suffix such as copy to get all names related to word that ends with copy
Get-command - verb "copy"

```
PS C:\Users\lenovo> get-command -verb "copy"

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Function        Copy-NetFirewallRule                               2.0.0.0    NetSecurity
Function        Copy-NetIPsecMainModeCryptoSet                     2.0.0.0    NetSecurity
Function        Copy-NetIPsecMainModeRule                          2.0.0.0    NetSecurity
Function        Copy-NetIPsecPhase1AuthSet                         2.0.0.0    NetSecurity
Function        Copy-NetIPsecPhase2AuthSet                         2.0.0.0    NetSecurity
Function        Copy-NetIPsecQuickModeCryptoSet                    2.0.0.0    NetSecurity
Function        Copy-NetIPsecRule                                  2.0.0.0    NetSecurity
Cmdlet          Copy-Item                                          3.1.0.0    Microsoft.PowerShell.Management
Cmdlet          Copy-ItemProperty                                  3.1.0.0    Microsoft.PowerShell.Management
```

To search a name in which the word comes in middle we use thee command
Get-command -name "*Notepad*"
Get command "*suspend*"

```
PS C:\Users\lenovo> get-command "*suspend*"

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Function        Suspend-BitLocker                                  1.0.0.0    BitLocker
Function        Suspend-PrintJob                                   1.1        PrintManagement
Function        Suspend-StorageBusDisk                             1.0.0.0    StorageBusCache
Cmdlet          Suspend-BitsTransfer                               2.0.0.0    BitsTransfer
Cmdlet          Suspend-Job                                        3.0.0.0    Microsoft.PowerShell.Core
Cmdlet          Suspend-Service                                    3.1.0.0    Microsoft.PowerShell.Management
```

For clear screen -use command -**cls**

Alias (2 names same meaning)

when we type **get-command -noun "service"** we get the service commands and command type as cmdlet

```
PS C:\Users\lenovo> get-command "*suspend*"

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Function        Suspend-BitLocker                                  1.0.0.0    BitLocker
Function        Suspend-PrintJob                                   1.1        PrintManagement
Function        Suspend-StorageBusDisk                             1.0.0.0    StorageBusCache
Cmdlet          Suspend-BitsTransfer                               2.0.0.0    BitsTransfer
Cmdlet          Suspend-Job                                        3.0.0.0    Microsoft.PowerShell.Core
Cmdlet          Suspend-Service                                    3.1.0.0    Microsoft.PowerShell.Management
```

to know cmdlet or command type is a alias - **get-command -commandtype "Alias"**

```
PS C:\Users\lenovo> get-command -commandtype "Alias"

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Alias           % -> ForEach-Object
Alias           ? -> Where-Object
Alias           ac -> Add-Content
Alias           Add-AppPackage                                     2.0.1.0    Appx
Alias           Add-AppPackageVolume                               2.0.1.0    Appx
Alias           Add-AppProvisionedPackage                          3.0        Dism
Alias           Add-ProvisionedAppPackage                          3.0        Dism
Alias           Add-ProvisionedAppxPackage                         3.0        Dism
Alias           Add-ProvisioningPackage                            3.0        Provisioning
Alias           Add-TrustedProvisioningCertificate                 3.0        Provisioning
Alias           algm ->                                            1.0.0.0    Microsoft.PowerShell.LocalAccounts
Alias           Apply-WindowsUnattend                              3.0        Dism
Alias           asnp -> Add-PSSnapin
Alias           blsmba ->                                          2.0.0.0    SmbShare
Alias           cat -> Get-Content
Alias           cd -> Set-Location
Alias           CFS -> ConvertFrom-String                          3.1.0.0    Microsoft.PowerShell.Utility
Alias           chdir -> Set-Location
Alias           clc -> Clear-Content
Alias           clear -> Clear-Host
Alias           clhy -> Clear-History
Alias           cli -> Clear-Item
Alias           clp -> Clear-ItemProperty
Alias           cls -> Clear-Host
Alias           clv -> Clear-Variable
Alias           cmpcfg ->                                          1.1        PSDesiredStateConfiguration
Alias           cnsn -> Connect-PSSession
Alias           compare -> Compare-Object
Alias           copy -> Copy-Item
Alias           cp -> Copy-Item
Alias           cpi -> Copy-Item
Alias           cpp -> Copy-ItemProperty
Alias           cssmbo ->                                          2.0.0.0    SmbShare
Alias           cssmbse ->                                         2.0.0.0    SmbShare
Alias           curl -> Invoke-WebRequest
Alias           cvpa -> Convert-Path
Alias           dbp -> Disable-PSBreakpoint
Alias           del -> Remove-Item
Alias           diff -> Compare-Object
Alias           dir -> Get-ChildItem
Alias           Disable-PhysicalDiskIndication                     2.0.0.0    Storage
Alias           Disable-StorageDiagnosticLog                       2.0.0.0    Storage
Alias           Dismount-AppPackageVolume                          2.0.1.0    Appx
Alias           dlu ->                                             1.0.0.0    Microsoft.PowerShell.LocalAccounts
```

Example of Alias
Clear -clear host
Clr -clear host
type - get-alias

Get your specific Alias using this command - **get-alias -Definition "clear-host"**

```
PS C:\Users\lenovo> get-alias -definition "clear-host"

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Alias           clear -> Clear-Host
Alias           cls -> Clear-Host
```

To get more info about alias - **get-command -name "get-alias" -syntax**

```
PS C:\Users\lenovo> get-command -name "get-alias" -syntax

Get-Alias [[-Name] <string[]>] [-Exclude <string[]>] [-Scope <string>] [<CommonParameters>]

Get-Alias [-Exclude <string[]>] [-Scope <string>] [-Definition <string[]>] [<CommonParameters>]
```